Practical 5

Aim: Programs on collections and LINQ

Collections:

Collections in C# are specialized classes used to store and manage groups of objects efficiently. Unlike arrays, collections are dynamic, meaning their size can grow or shrink at runtime. They provide built-in methods for adding, removing, searching, sorting, and accessing elements, making them ideal for handling data in real-world applications. The System.Collections namespace contains non-generic collections, while the System.Collections.Generic namespace provides generic collections that are type-safe and offer better performance. Common collections include List<T>, Stack<T>, Queue<T>, Dictionary<TKey, TValue>, HashSet<T>, and LinkedList<T>.

LINQ

Language Integrated Query (LINQ) is a feature introduced in C# that allows developers to query and manipulate data directly within the programming language using a consistent syntax. LINQ can retrieve data from arrays, collections, databases, XML documents, and other data sources without requiring different query languages. It improves code readability, reduces complexity, and provides compile-time type checking. LINQ supports both **Query Syntax** (similar to SQL) and **Method Syntax** (using extension methods).

Programs:

5.1. List Collection

```
using System;
using System.Collections.Generic;

namespace CollectionsDemo
{
   class Program
   {
      static void Main(string[] args)
      {
         List<string> students = new List<string>();
         students.Add("Rahul");
         students.Add("Amit");
         students.Add("Priya");
         students.Add("Neha");
         Console.WriteLine("Student List");
         foreach (string s in students)
         {
            Console.WriteLine(s);
         }
         Console.ReadKey();
      }
```

```
    }
}
```

5.2 Stack Collection

```
using System;
using System.Collections;

namespace CollectionsDemo
{
    class Program
    {
        static void Main(string[] args)
        {
            Stack stack = new Stack();
            stack.Push("Apple");
            stack.Push("Banana");
            stack.Push("Orange");
            Console.WriteLine("Removed Item : " + stack.Pop());
            Console.WriteLine("Remaining Items");
            foreach (object item in stack)
            {
                Console.WriteLine(item);
            }
            Console.ReadKey();
        }
    }
}
```

5.3 Queue Collection

```
using System;
using System.Collections;

namespace CollectionsDemo
{
    class Program
    {
        static void Main(string[] args)
        {
            Queue q = new Queue();
            q.Enqueue("Student1");
            q.Enqueue("Student2");
            q.Enqueue("Student3");
            Console.WriteLine("Dequeued : " + q.Dequeue());
            Console.WriteLine("Remaining Students");
            foreach (object item in q)
            {
                Console.WriteLine(item);
            }
            Console.ReadKey();
        }
```

```
    }
}
```

5.4 Dictionary Collection

```
using System;
using System.Collections.Generic;

namespace CollectionsDemo
{
    class Program
    {
        static void Main(string[] args)
        {
            Dictionary<int, string> students = new Dictionary<int, string>();
            students.Add(101, "Rahul");
            students.Add(102, "Neha");
            students.Add(103, "Amit");
            foreach (KeyValuePair<int, string> item in students)
            {
                Console.WriteLine(item.Key + " " + item.Value);
            }
            Console.ReadKey();
        }
    }
}
```

5.5 HashSet Collection

```
using System;
using System.Collections.Generic;

namespace CollectionsDemo
{
    class Program
    {
        static void Main(string[] args)
        {
            HashSet<int> numbers = new HashSet<int>();
            numbers.Add(10);
            numbers.Add(20);
            numbers.Add(30);
            numbers.Add(20);
            foreach (int n in numbers)
            {
                Console.WriteLine(n);
            }
            Console.ReadKey();
        }
    }
}
```

5.6 LINQ Where (Filter Data)

```
using System;
using System.Linq;
namespace LINQDemo
{
   class Program
   {
      static void Main(string[] args)
      {
         int[] numbers = { 12, 45, 20, 5, 80, 32 };
         var result = from n in numbers
                      where n > 20
                      select n;
         Console.WriteLine("Numbers Greater Than 20");
         foreach (int n in result)
         {
            Console.WriteLine(n);
         }
         Console.ReadKey();
      }
   }
}
```

5.7 LINQ OrderBy

```
using System;
using System.Linq;

namespace LINQDemo
{
   class Program
   {
      static void Main(string[] args)
      {
         int[] marks = { 65, 40, 95, 80, 70 };
         var result = from m in marks
                      orderby m
                      select m;
         foreach (int m in result)
         {
            Console.WriteLine(m);
         }
         Console.ReadKey();
      }
   }
}
```

5.8 LINQ Select (Projection)

```
using System;
using System.Linq;

namespace LINQDemo
{
   class Program
   {
```

```
        static void Main(string[] args)
        {
            string[] names = { "Rahul", "Amit", "Neha" };
            var upper = names.Select(x => x.ToUpper());
            foreach (string s in upper)
            {
                Console.WriteLine(s);
            }
            Console.ReadKey();
        }
    }
}
```

5.9 LINQ GroupBy

```
using System;
using System.Linq;
class Student
{
    public string Name { get; set; }
    public string Department { get; set; }
}
class Program
{
    static void Main()
    {
        Student[] students =
        {
            new Student{Name="Rahul",Department="IT"},
            new Student{Name="Neha",Department="CS"},
            new Student{Name="Amit",Department="IT"},
            new Student{Name="Priya",Department="CS"}
        };
        var groups = students.GroupBy(x => x.Department);
        foreach (var g in groups)
        {
            Console.WriteLine(g.Key);

            foreach (var s in g)
            {
                Console.WriteLine(s.Name);
            }
            Console.WriteLine();
        }

        Console.ReadKey();
    }
}
```

5.10 LINQ Count, Max and Average

```
using System;
using System.Linq;

namespace LINQDemo
```

```
{
   class Program
   {
      static void Main(string[] args)
      {
         int[] marks = { 78, 45, 90, 66, 82 };
         Console.WriteLine("Count = " + marks.Count());
         Console.WriteLine("Maximum = " + marks.Max());
         Console.WriteLine("Minimum = " + marks.Min());
         Console.WriteLine("Average = " + marks.Average());
         Console.ReadKey();
      }
   }
}
```

5.11 LINQ First and Last

```
using System;
using System.Linq;
namespace LINQDemo
{
   class Program
   {
      static void Main()
      {
         int[] numbers = { 10, 20, 30, 40, 50 };
         Console.WriteLine("First : " + numbers.First());
         Console.WriteLine("Last : " + numbers.Last());
         Console.ReadKey();
      }
   }
}
```

5.12 LINQ Join

```
using System;
using System.Linq;
class Student
{
   public int Id;
   public string Name;
}
class Result
{
   public int Id;
   public int Marks;
}
class Program
{
   static void Main()
   {
      Student[] students =
      {
         new Student{Id=1,Name="Rahul"},
         new Student{Id=2,Name="Neha"},
```

```
            new Student{Id=3,Name="Amit"}
        };
        Result[] results =
        {
            new Result{Id=1,Marks=85},
            new Result{Id=2,Marks=90},
            new Result{Id=3,Marks=80}
        };
        var query = from s in students
                    join r in results
                    on s.Id equals r.Id
                    select new
                    {
                        s.Name,
                        r.Marks
                    };
        foreach (var item in query)
        {
            Console.WriteLine(item.Name + " " + item.Marks);
        }
        Console.ReadKey();
    }
}
```